1.3 Exponential Models

Definitions

Definition 1 (Compound Interest). *Suppose a principal P earns interest at the annual rate of r, and interest is compounded m times a year. Then F after t years is*

$$F = P(1+i)^n = P(1+\frac{r}{m})^{mt}$$

where $n = mt$=number of time periods and $i = r/m$ =interest per period.

Definition 2 (Present Value). *Given $F = P(1+r/m)^{mt}$ and say we want to know how many dollars P to set aside so we will have a future amount of F dollars after t years. We solve for P:*

$$P = \frac{F}{(1+r/m)^{mt}} = \textit{Present value}$$

Definition 3. *If a principal P earns interest at an annual rate of r, and interest is compounded continuously, then F after t years is $F = Pe^{rt}$.*

Definition 4 (Present Value for Continuous Compounding). *Present value for continuous compounding is given by $P = Fe^{-rt}$ = Present value.*

1.5 Logarithms

Definitions

Definition 5 (Logarithm)**.** *Let $a > 0$ and $a \neq 1$. If $x > 0$, then the logarithm base a of x, denoted $\log_a x$ is defined as follows:*

$$y = log_a x \textit{ if and only if } x = a^y$$

Definition 6 (Common Logarithm/Log Base 10)**.** *Let $a = 10$, then*

$$y = \log_{10} x = \log(x) \textit{ if and only if } x = 10^y$$

Definition 7 (Natural Log/Log Base e)**.** *Let $a = e$, then*

$$y = \log_e x = \ln(x) \textit{ if and only if } x = e^y$$

Properties of Logarithms

1. $a^{\log_a x} = x$ if $x > 0$ (so exponentiation undoes the logarithm).
2. $\log_a a^x = x$ for all x (so logarithm undoes exponentiation).
3. $\log_a xy = \log_a x + \log_a y$
4. $\log_a x/y = \log_a x - \log_a y$
5. $\log_a x^c = c \log_a x$
6. Let $a > 0$, then $a^x = (a)^x = (e^{\ln a})^x = e^{x \ln a}$
7. Change of Base: $\log_a x = \frac{\log_b x}{\log_b a}$.